Infosys - Coding 1

HackerRank

A frog starts at position 0 and must reach position N. At each step, it may jump 1 or 2 positions forward. Landing on any position i (where i > 0) costs a[i]. Position 0 is the starting point and is considered free. If the frog jumps 2 positions in a single leap (e.g., from i to i+2), it pays an additional long-jump fee F. This fee is in addition to the landing cost a[i+2]. Find the minimum total cost to reach position N.

Case 1:

Input:

3 10 1 2 3

Output:

6

Explanation:

The path 0 → 1 → 2 → 3 involves only single jumps, incurring landing costs of 1, 2, and 3 respectively, for a total of 6. Any path utilizing a long jump would add the fee F=10, making it more expensive.

Case 2:

Input:

4 0 10 1 10 1

Output:

2

Explanation:

With F=0, long jumps are free. Best path: 0 → 2 → 4, landing on positions 2 and 4 for costs 1 + 1 = 2.

Input Format

The first line contains an integer, n, denoting the target position the frog must reach. The second line contains an integer, F, denoting the additional cost for a long jump. The third line contains N space-separated integers a[i].

Constraints

1 <= n <= 10^5 1 <= F <= 10^5 -10^9 <= a[i] <= 10^9

Output Format

The output should be an integer value

Sample Input 0

```
3
10
1 2 3
```

Sample Output 0

```
6
```

Explanation 0

The path 0 → 1 → 2 → 3 involves only single jumps, incurring landing costs of 1, 2, and 3 respectively, for a total of 6. Any path utilizing a long jump would add the fee F=10, making it more expensive.

Sample Input 1

```
4
0
10 1 10 1
```

Sample Output 1

```
2
```

Explanation 1

With F=0, long jumps are free. Best path: 0 → 2 → 4, landing on positions 2 and 4 for costs 1 + 1 = 2.